MATTHIJS MOERKERKE

Postdoctoral researcher

ABOUT

°15/06/1992, Brugge
Belgian
In possession of driving licence

CONTACT

Work

University Ghent (80%)
Dept. Rehabilitation Sciences - Spine, Head and Pain Research Unit.
University Hospital Ghent,
Corneel Heymanslaan 10, B3, room 013
9000, Ghent, Belgium.
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University Leuven
Dept. Rehabilitation Sciences - Neurorehabilitation Research Group
Tervuursevest 101, 3001 Leuven
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Personal

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LANGUAGES

Dutch: native language - C2
English: very fluently - C1
French: basic - A2
Spanish: basic - A1

EDUCATION

2017-2022 **PhD Research:** The role of oxytocin in autism - a randomized clinical trial
Center for Developmental Psychiatry, KU Leuven

2016-2017 **Postgraduate Studies** in Advanced Medical Imaging - summa cum laude
Center for Developmental Psychiatry, KU Leuven

2015-2016 **Master of Biomedical Sciences** minor clinical - cum laude
Center for Developmental Psychiatry, KU Leuven

2010-2014 **Bachelor of Biomedical Sciences**
KU Leuven Department of Kortrijk (KULAK)

ADDITIONAL TRAINING

2025 **Research stay at Stanford University, CA, USA**
Pediatric pain research lab of Prof Laura Simons - (Faculty and FWO travel grant)

2023 **Research stay at the University of Oslo, Norway**
Oxytocin research lab of Prof Daniel Quintana - (FWO travel grant)

2022 **PhD in Neuroscience Certificate**
Leuven Brain Institute - KU Leuven, Belgium

2022 **Oxytocin conference and workshop - invited talk**
Ettore Majorana Foundation, Erice, Sicily, Italy - (FWO travel grant)

2021 **Regulatory medical writing training programme**
Janssen Pharmaceutica NV - KU Leuven, Belgium

2019 **Freesurfer - MRI analysis tool**
Martinos Center, MIT-Harvard, Boston, USA - (FWO travel grant)

2019 **Good Clinical (Research) Practice (GCP) Certificate**

2017 **Spanish language classes**
Centrum voor Levende Talen (CLT), KU Leuven, Belgium

2017 **Certificate MATLAB and Octave**
École polytechnique fédérale de Lausanne

2014 **Certificate of laboratory animal science category B (FELASA B)**

SKILLS & ACQUIRED TECHNIQUES

Medical imaging proficiencies

Scanning, pre-processing and analysis of images taken with Magnetic Resonance Imaging or **MRI** (uni- and multivariate functional, structural and diffusion weighted MRI) and electroencephalograpy or **EEG** (including stress physiology measures: heart rate, respiration rate and skin conductance); using medical imaging software including Letswave 6 (https://www.letswave.org/), SPM 8-12 (Wellcome Trust Centre for Neuroimaging, Londen, UK), CAT 12 (Jena, Germany) and FreeSurfer (The Martinos Center, Boston, USA).

Programming skills

Programming scripts for fMRI, statistical and behavioural experiments in MATLAB, RStudio and Python.

Neurocognitive expertise

Experimental design and development, cognitive and diagnostic testing.

Laboratorial skills

Animal experiments, organ dissection, tissue embedding, microscopic evaluation and histological analysis.

Clinical skills

Placing infuses, drawing blood, intramuscular injection, subdermal injection, performing ECG/EEG, CPR training.

TEACHING AND MENTORING ROLES

Teaching: *Evidence based medicine and statistics*, 3rd BA Rehabilitation sciences, UGent

- **Lecturer** of theoretical EBM classes & coordinator of practical EBM seminars

Co-promotor PhD thesis: 2 PhD candidates

Promotor MSc thesis: 19 students (Rehabilitation sciences, Biomedical Sciences, Medicine)

Co-pomotor MSc thesis: 12 students (Rehabilitation sciences, Biomedical Sciences)

Daily supervisor: 24 students

Jurymember PhD thesis: 1 PhD candidate

Jurymember MSc thesis: 10 students

OTHER ACTIVITIES

2025 **Scientific program and award manager** Pain Science in Motion conference 2026
2024 Scientific outreach in mainstream media, including EOS: www.eoswetenschap.eu
2023 Scientific outreach via ACAMH papers podcast: https://bit.ly/3t6z9tr
2020-current Tutoring anatomy and biology to bachelor students (Bijleshuis)
2017-current Yearly workshops during 'Dag van de Wetenschap' (Day of Science)
2017-2023 Backup ombudsman for the Postgraduate Studies in Advanced Medical Imaging

Memberships

2025-current Active Monitoring of Oxytocin Research Evidence (AMORE)
Expert panel member - University of Oslo
2025-current Alliantieonderzoeksgroep UGent - VUB: Motor Mind
2024-current International Association for the Study of Pain (IASP)
2023-current **Pain In Motion (PIM), UGent - active Steering Group member**
2022-current The Belgian Association for Psychological Sciences (BAPS)
2021-current The KU Leuven Child and Youth Institute (L-C&Y), KU Leuven
2019-current The Leuven Brain Institute (LBI), KU Leuven
2018-current The International Society for Autism Research (INSAR)
2017-current Leuven Autism Research (LAuRes), KU Leuven

Flexi work at JABCO (jabco.be): Organizing scientific symposia and webinars. Creating logo's, websites and webinars for scientific societies (e.g. Belgian Infection Control Society)

PEER REVIEWED PUBLICATIONS

Moerkerke, M., Timmers, I., Coppieters, I., Van Oosterwijck, J. (2025). Oxytocin in chronic pain: from analgesic to biopsychosocial adjuvant - an opinion paper. In preparation.

Vyverman, J., De Baere, R. (joint first author), Timmers, I., Coppieters, I., Van Oosterwijck, J. **Moerkerke, M.** (2025). The Stress-Pain Connection in Chronic Primary Pain: A Systematic Review and Meta-Analysis of Physiological Stress Markers in Relation to Experimental Pain Responses. ***MedRxiv***, 10.1101/2025.09.24.25336431. And under review in ***Clinical Psychology Review*** (IF: 12.8)

Cnockaert, E., Cagnie, B., Meeus, M., Chys, M., Steverlynck, C., **Moerkerke, M.,** Van Oosterwijck, J., (2025) Human Assumed Central Sensitization in Interictal Migraine: a Systematic Review and Meta-Analysis. Under review in ***Pain Practice*** (IF: 2.7)

Van Oosterwijck, S., Wijnen, J., Brandt, M., Billens, A., Meirezonne, H., Dhondt, E., Bernaers, L., De Greef, I., Jeurissen, B., Willems, T., Matheve, T., Hodges, P., **Moerkerke, M.,** Van Oosterwijck, J., Danneels, L., (2025). Efficacy of specific skilled motor versus general exercise training on peripheral muscle and central neural alterations in recurrent low back pain: protocol of a randomized clinical trial. Under review in ***European Journal of Physiotherapy*** (IF: 2.284)

Iversen, I. A., Alaerts, K., Bakermans-Kranenburg, M., Becker, B., Blair, J., Bartz, J., Connelly, J., Ditzen, B., Ebner, N., Kang, H., Lawson, E., Lønfeldt, N., **Moerkerke, M.**,... Quintana, D. (2025). The active monitoring of oxytocin research evidence (AMORE) platform. ***Open Science Framework*** https://doi.org/10.31234/osf.io/zxdwn_v1

Deilhaug, E., **Moerkerke, M.,** Sartorius, A.I., Kang, H., Smith-Meyer Kilda E., Walle, K.M., Elvsåshagen, T., Westlye, L.T., Nærland, T., Andreassen, O.A., Quintanan D.S. (2025). Using EEG to measure the neural effects of oxytocin administration: A meta-analysis and systematic review. ***MedRxiv***, 2025.08.25.25334355

Moerkerke, M., Vyverman, J., De Baere, R., Timmers, I., Coppieters, I., Van Oosterwijck, J (2025). A comprehensive mapping of stress system interactions with pain and their contribution to extent and chronification of musculoskeletal pain: a study protocol. ***PLoS ONE***, 20(6), e0324089. https://doi.org/10.1371/journal.pone.0324089 (IF: 2.9).

Tang, T., **Moerkerke, M.,** Daniels, N., Van der Donck, S., Steyaert, J., Naulaers, G., . . . Boets, B. (2025). Face Processing in School-Aged Preterm Children: Assessing Neural Sensitivity to Facial Identity and Expression Using Frequency-Tagging EEG. ***Biological Psychiatry: Cognitive Neuroscience and Neuroimaging.*** doi:10.1016/j.bpsc.2025.05.015 (IF: 4.8)

Moerkerke, M., Daniels, N. (joint first author), Van der Donck, S., Tang, T., Prinsen, J., Yargholi, E., Steyaert, J., Alaerts, K., Boets, B. (2024). Impact of chronic intranasal oxytocin administration on face expression processing in autistic children: a randomized controlled trial using fMRI. ***Molecular Autism,*** *15, 53 (2024). https://doi.org/10.1186/s13229-024-00635-z* (IF: 6.503)

Tang, T., Pledts, K., **Moerkerke, M.,** Van der Donck, S., Bollen, B., Steyaert, J., Alaerts, K., Ortibus, E., Naulaers, G., Boets, B. (2024). Face Processing in Prematurely Born Individuals—A Systematic Review. ***Brain Sciences***, doi:10.3390/brainsci14121168 (IF: 2.7)

Evenepoel, M., Daniels, N. (joint first author), **Moerkerke, M.,** Van de Vliet, M., Prinsen, J., Tuerlinckx, E., Steyaert, J., Boets, B., Alaerts, K. Joossens, M. (2024). Oral microbiota in autistic children: Diagnosis-related differences and associations with clinical characteristics. ***Brain, Behavior, & Immunity - Health***, 38. doi:10.1016/j.bbih.2024.100801 (IF: 4.872)

Moerkerke, M., Daniels, N. (joint first author), Tibermont, L., Tang, T., Evenepoel, M., Van der Donck, S., Debbaut, E., Bamps, A., Prinsen, J., Chubar, V., Claes, S., Vanaudenaerde, B., Willems, L., Steyaert, J., Boets, B., Alaerts, K. (2024). Chronic oxytocin administration stimulates the oxytocinergic system: an RCT in autistic children. ***Nature Communications***, doi: https://doi.org/10.1038/s41467-023-44334-4 (**IF: 17.694**)

Alaerts, K., **Moerkerke, M. (joint first author)**, Daniels, N., Qianqian, Z., Steyaert, J., Prinsen, J., Bart, B., (2024). Chronic oxytocin improves neural decoupling at rest in children with autism: An exploratory RCT. In press in ***Journal of Child Psychology and Psychiatry***, doi: https://doi.org/10.1111/jcpp.13966 (IF: 8.265)

Deferm, W., Tang, T. (joint first author), **Moerkerke, M.,** Daniels, N., Steyaert, J., Alaerts, K., Ortibus, E., Naulaerts, G., Boets, B. (2024). Subtle Microstructural Alterations in White Matter Tracts Involved in Socio-emotional Processing After Very Preterm Birth. ***NeuroImage: Clinical***, doi: 10.1016/j.nicl.2024.103580 (IF: 4.891)

Alaerts, K., Daniels, N., **Moerkerke, M. (joint first author),** Evenepoel, M., Tang, T., Van der Donck, S., Chubar, V., Claes, S., Steyaert, J., Boets, B., & Prinsen, J. (2023). At the head and heart of oxytocin's stress-regulatory neural and cardiac effects: a chronic administration RCT in children with autism. ***Psychotherapy and Psychosomatics***, doi: https://doi.org/10.1159/000534114 (IF: 25.617)

Evenepoel, M., **Moerkerke, M. (joint first author),** Daniels, N., Chubar, V., Claes, S., Turner, J., Vanaudenaerde, B., Willems, L., Verhaeghe, J., Prinsen, J., Steyaert, J., Boets, B., & Alaerts, K. (2023). Endogenous oxytocin levels in children with autism: Associations with cortisol levels and oxytocin receptor gene methylation. ***Translational Psychiatry*** 2023 13:1, 13(1), 1–9. doi: https://doi.org/10.1038/s41398-023-02524-0 (IF: 7.989)

Moerkerke, M., Daniels, N. (joint first author), Van der Donck, S., Tibermont, L., Tang, T., Debbaut, E., Bamps, A., Prinsen, J., Steyaert, J., Alaerts, K., Boets, B. (2023). Can repeated intranasal oxytocin administration affect reduced neural sensitivity towards expressive faces in autism? A randomized controlled trial. ***Journal of Child Psychology and Psychiatry***. doi: https://doi.org/10.1111/jcpp.13850 (IF: 8.265)

Daniels, N., **Moerkerke, M. (joint first author),** Steyaert, J., Bamps, A., Debbaut, E., Prinsen, J., Tang, T., Van der Donck, S., Boets, B., Alaerts, K. (2023). Effects of multiple-dose intranasal oxytocin administration on social responsiveness in children with autism: A randomized, placebo-controlled trial. ***Molecular Autism***, 14, 16. doi: https://doi.org/10.1186/s13229-023-00546-5 (IF: 6.503)

Qiao, Z., Van der Donck, S. (joint first author), **Moerkerke, M.,** Dlhosova, T., Vettori, S., Dzhelyova, M.,Van Winkel, R., Alaerts, K., Boets, B. (2022). Frequency-tagging EEG of superimposed social and non-social visual stimulation streams provides no support for social salience enhancement after intranasal oxytocin administration. ***Brain Sciences*** 2022, Vol. 12, Page 1224, 12(9), 1224. doi: https://doi.org/10.3390/brainsci12091224 (IF: 3.333)

Van der Donck,S., **Moerkerke, M.,** Dlhosova, T., Vettori, S., Dzhelyova, M., Alaerts, K., & Boets, B. (2022). Monitoring the effect of oxytocin on the neural sensitivity to emotional faces via frequency-tagging EEG: A double-blind, cross-over study. ***Psychophysiology***. doi: https://doi.org/10.1111/psyp.14026 (IF: 4.348)

Moerkerke, M., Peeters,M., de Vries, L., Daniels,N., Steyaert, J., Alaerts, K., & Boets,B. (2021). Endogenous Oxytocin Levelsin Autism-A Meta-Analysis.. ***Brain Sciences***,11(11). doi: https://doi.org/10.3390/brainsci11111545 (IF: 3.333)

Van Dessel, J., Sonuga-Barke, E. J. S., **Moerkerke, M.,** Vander Oord, S., Morsink, S., Lemiere, J., & Danckaerts, M. (2021). The Limits of Motivational Influence in ADHD: No Evidence for an Altered Reactionto Negative Reinforcement.. ***Social Cognitive and Affective Neuroscience*** (SCAN). doi: https://doi.org/10.1093/scan/nsab111 (IF: 4.235)

Moerkerke, M., Bonte, M. -L. (joint first author), Daniels,N., Chubar, V., Alaerts, K., Steyaert, J., & Boets, B. (2021). Oxytocin receptor gene (OXTR) DNA methylation is associated with autism and related social traits-A systematic review. ***Research In Autism Spectrum Disorders***, 85, 14 pages. doi: https://doi.org/10.1016/j.rasd.2021.101785 (IF: 3.293)

Van Dessel, J., Danckaerts, M., **Moerkerke, M.**, Van der Oord, S., Morsink, S., Lemiere, J., & Sonuga-Barke, E. (2021). Dissociating brain systems that respond to contingency and valence during monetary loss avoidance in adolescence. ***Brain And Cognition***, 150, 1- 9. doi: https://doi.org/10.1016/j.bandc.2021.105723 (IF: 2.682)

Van Dessel, J., Sonuga-Barke, E., **Moerkerke, M.,** Van der Oord, S., Lemiere, J., Morsink, S., & Danckaerts, M. (2019). The amygdala in adolescents with attentiondeficit/ hyperactivity disorder: Structural and functional correlates of delay aversion. ***World Journal of Biological Psychiatry***, 1-12. doi: https://doi.org/10.1080/15622975.2019.1585946 (IF: 3.418)

Van Dessel,J., Morsink, S., Van der Oord, S., Lemiere, J., **Moerkerke, M.,** Grandelis, Danckaerts,M. (2018). Waiting impulsivity: a distinctive feature of ADHD neuropsychology?. ***Child Neuropsychology***, 25(1), 122-129.
doi: https://doi.org/10.1080/09297049.2018.1441819 (IF: 2.597)

AWARDS

PhD Dissertation Award 2024, awarded by the International Society for Autism Research (INSAR), presented at the Annual meeting in Melbourne, Australia: https://www.autism-insar.org/page/RecognitionAwards

The "Belgium's Got Talent" Prize 2023, awarded by the Belgian College of Neuropsychopharmacology and Biological Psychiatry, for best PhD thesis: https://www.bcnbp.org/belgiums-got-talent

The 2023 Thérèse Vandereecken, spouse Boes, Adrienne Boes, and Viviane Boes Prize for Clinical Medicine (Royal Academy of Medicine of Belgium, €7,000) recognized our research on oxytocin's neural, biological, and behavioral effects in children with autism, central to my PhD, awarded to Prof. Kaat Alaerts (co-promotor).

2nd Runners up for best oral presentation: Joren Vyverman, Robrecht De Baere, Inge Timmers, Iris Coppieters, Jessica Van Oosterwijck, **Matthijs Moerkerke** (2025). Stress and pain systems: A meta-analysis of their complex physiological interplay. Oral presentation presented by J.V. at the Belgian Pain Society, 22 March 2024, Antwerp, Belgium (abstract book 12 - p. 20).

Best poster award: De Baere R, Vyverman R, Coppieters I, Van Oosterwijck J, **Moerkerke M,** Timmers I. Stress in pain (strain) protocol: examining neural correlates of pain-stress interactions. Poster presented by R.D. at the closed Pain Research Meeting (PRM) 2024, Azur en Ardenne, Barvaux-sur-Ourthe, Belgium, September 16th 2024 (poster presentation)

Best poster award: Nataliia Grytsenko, **Matthijs Moerkerke,** Nicky Daniels, Stephanie Van der Donck, Laura Tibermont, Tiffany Tang, Jean Steyaert, Kaat Alaerts, Bart Boets (2023). Effects of Chronic Intranasal Oxytocin Administration on Neural Sensitivity for Face Identity Processing in Children with Autism: an RCT. Presented by N.G. at Belgian Association for Psychological Sciences (BAPS), May 25-26 2023 (poster presentation).

First price in the clinical neuroscience session at the ECNP Workshop on Neuropsychopharmacology for Junior Scientists in Europe, 15-18 March 2018, Nice, France

SUMMARY OF PRESENTATIONS AT CONFERENCES

Moerkerke, M., The role of oxytocin in modulating pain and its interaction with stress in the context of musculoskeletal pain. EFIC 2025, Lyon, France. (oral talk)

Moerkerke, M., PhD Dissertation Award Presentation: The role of oxytocin in autism. International Society for Autism Research (INSAR) 2024, Melbourne, Australia. (oral talk)

Moerkerke, M., Daniels, N., Tibermont, L., Van der Donck, S., Tang, T., Prinsen, J., . . . Boets, B. (2023). Chronic oxytocin administration does not rescue reduced neural sensitivity towards emotional faces in autism. In NeuroCog 2023. Brussels, Belgium. (oral talk)

Moerkerke, M., Daniels, N., Tibermont, L., Van der Donck, S., Tang, T., Prinsen, J., Bamps A., Debbaut E., Steyaert, J., Alaerts, K., Boets, B. (2023). Influence of repeated intranasal oxytocin administration on neural activity during face processing in children with autism. – International Society for Autism Research (INSAR), Stockholm, Sweden (poster presentation)

Moerkerke, M., Tibermont, L., Van der Donck, S., Daniels, N., Tang, T., Prinsen, J., Steyaert, J., Alaerts, K., Boets, B. (2022). Can longterm intranasal oxytocin administration rescue reduced neural sensitivity towards emotional faces in autism? Presented at the OXYTOCIN AND VASOPRESSIN. FROM BRAIN MODULATION, TO EPIGENETIC REGULATION AND CLINICAL APPLICATIONS, Erice, Sicily, Italy, 29 May 2022-02 Jun 2022 (invited oral talk)

Moerkerke, M., Daniels, N., Chubar, V., Van der Donck, S., Tang, T., Prinsen, J., Claes, S., Steyaert, J., Alaerts, K., Boets, B. (2022). Oxytocin system alterations in autism? Assessing endogenous oxytocin levels and OXTR epigenetics throughout a long-term intranasal oxytocin treatment. Presented at the OXYTOCIN AND VASOPRESSIN. FROM BRAIN MODULATION, TO EPIGENETIC REGULATION AND CLINICAL APPLICATIONS, Erice, Sicily, Italy, 29 May 2022-02 Jun 2022 (poster presentation)

Moerkerke M., Chubar V., Daniels N., Van der Donck S., Tang T., Tibermont L., Steyaert J., Alaerts K., Boets B. (2022). Oxytocin receptor Epigenetics in Autism Spectrum Disorders – KU Leuven Child & Youth Institute Leuven, Belgium (oral talk)

Moerkerke M., Tibermont L., Van der Donck S., Daniels N., Steyaert J., Alaerts K., Boets B. (2021). The effect of oxytocin on reduced neural sensitivity towards emotional faces in autism – Leuven Brain Institute (LBI), KU Leuven, Belgium (poster presentation)

Moerkerke M., Daniels N., Van der Donck S., Steyaert J., Alaerts K., Boets B. (2021). Validating an EEG-based biomarker of socio-communicative impairments in ASD – International Society for Autism Research (INSAR) virtual (poster presentation)

Daniels, N., **Moerkerke, M.**, Prinsen, J., Boets, B., Steyaert, J., Alaerts, K. (2021). Measurement of Sympathetically Driven Autonomic Arousal during Eye Contact in Children with and without an Autism Diagnosis. Presented at the International Society for Autism Research (INSAR), virtual, 03 May 2021-07 May 2021. Open Access

Moerkerke, M., Daniels, N., Van der Donck, S., Steyaert, J., Alaerts, K., Boets, B. (2021). Neurobiological marker and intervention for socio-communicative impairments in autism spectrum disorders. In: *EUROPEAN NEUROPSYCHOPHARMACOLOGY*: *vol. 44*, (S51-S52). doi: 10.1016/j.euroneuro.2021.01.079 (poster presentation)

Moerkerke, M., Daniels, N., Stuer-Jespers, A., Steyaert, J., Alaerts, K., Boets, B. (2018). In search for an innovative neural marker and intervention for socio-communicative difficulties in children with and without autism spectrum disorders. (Abstract No. 24). Presented at the Understanding the Neuroregulatory Actions of Oxytocin and its Potential Clinical Applications, Erice, Sicily, Italy, 22 May 2018-27 May 2018. Open Access (poster presentation)